úvod

$\max\{f(x) \mid x \in X\}$ $\quad f: \mathbb{R}^n \to \mathbb{R},\ X \subseteq \mathbb{R}^n$

↑ množina přípustných řešení

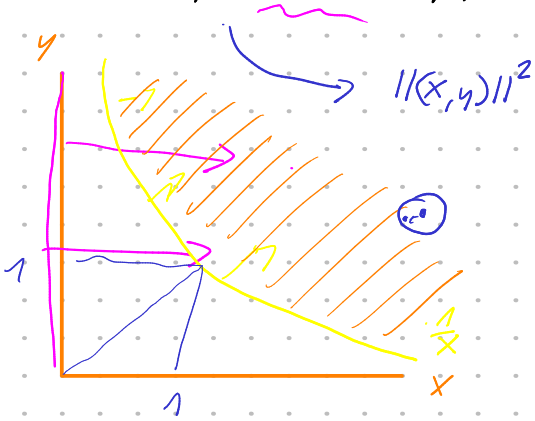
řešení je $\underset{x \in X}{\arg\max}\, f(x)$ → množina všech řešení

arg max → řešení tedy x

max → hodnota tedy $f(x)$

1.1

a) $\min\{x^2+y^2 \mid x>0,\ xy \geq 1\}$



→ $\|(x,y)\|^2$

Graficky:

$\arg\min = \{1, 1\} \to \min = 2$

Analýzou:

$y = \frac{1}{x} \to f(x) = x^2 + \frac{1}{x^2}$

$f'(x) = 2x - 2\frac{1}{x^3} = 0 \to x^4 - 1 = 0$

$x^4 = 1$

$x = \pm 1 \to x = 1$

$y = \frac{1}{1}$

$\lim_{x \to 0} f(x) = \infty$, $\lim_{x \to \infty} f(x) = \infty$ $\Big\} \Rightarrow (1,1) = \arg\min f(x)$

c) $\min\{x \mid x \in \mathbb{R},\ x \geq a_i,\ \forall i \in 1, \dots, n\},\ a_i \in \mathbb{R}$

$\underset{x \in X}{\arg\min}\, x = \{\max a_i\} \to \min = \max a_i$

f)

$S(h,r) = \pi r^2 + 2\pi r h$

$V(h,r) = \pi r^2 h = 0{,}5\ dm^3$

$\mapsto \min\{\pi r^2 + 2\pi r h \mid \pi r^2 h = 1/2,\ r>0,\ h>0\}$

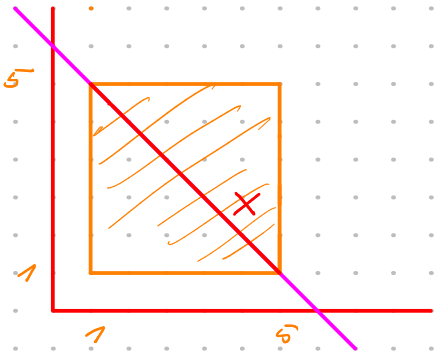
$\hookrightarrow h = \frac{1}{2\pi r^2}$

$f = \pi r^2 + 2\pi r \frac{1}{2\pi r^2} = \pi r^2 + \frac{1}{r}$

$f' = 2\pi r - \frac{1}{r^2} = 0 \to 2\pi r^3 = 1 \to r^3 = \frac{1}{2\pi} \to r = \sqrt[3]{\frac{1}{2\pi}} \to h = \frac{1}{2\pi\left(\sqrt[3]{\frac{1}{2\pi}}\right)^2} = \frac{1}{\sqrt[3]{2\pi}}$

$\arg\min = \{(\frac{1}{\sqrt[3]{2\pi}}, \frac{1}{\sqrt[3]{2\pi}})\}$

k) $x \in \langle 1;5\rangle, y \in \langle 1;5\rangle, x+y=6$



$\max\{xy^2 \mid 1 \le x \le 5,\ 1 \le y \le 5,\ x+y=6\}$

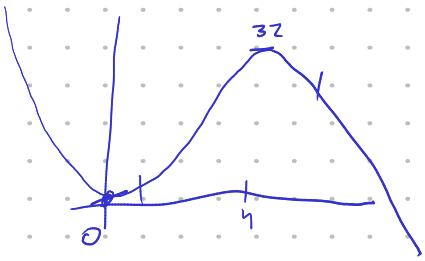
$\hookrightarrow x = 6-y$

$f = (6-y)y^2 = 6y^2 - y^3$

$f' = 12y - 3y^2 = 3y(4-y) = 0 \to y \in \{0, 4\}$

ne podmínky $\to x = 2$

$f(0) = 0,\ f(4) = 32$



$\arg\max = \{(2,4)\} \to \max = 32$

$\arg\min = \{(5,1)\} \to \min = 5$

i) $y = x^2$, nejblíže $(3,0)$

$\min\{\sqrt{(x-3)^2 + y^2} \mid x^2 = y\}$

$\hookrightarrow f = (x-3)^2 + y^2 \to h = (x-3)^2 + x^4$

$h' = 2(x-3) + 4x^3 = 0 \to x - 3 + 2x^3 = 0$

$(x-1)(x^2+2x+3) = 0 \to x = 1 \to y = 1$

$\arg\min = \{(1,1)\}$

$\min = \sqrt{5}$

(1.2) $X \in \mathbb{R}^n$, $f: \mathbb{R}^n \to \mathbb{R}$, $g: \mathbb{R} \to \mathbb{R}$ rostoucí

Pak $z \in \arg\min_{x \in X} f(x) = \arg\min_{x \in X} g(f(x))$

$f(z) \le f(x)\ \forall x \in X$

$g(f(z)) \le g(f(x))\ \forall x \in X$

(1.1)

b) $x_1, \dots, x_n$, $x_i \in \{-1;1\}$

$x_1 \cdot x_2 \cdot \dots \cdot x_n > 0$

$x_1 + x_2 + \dots + x_n$ min!

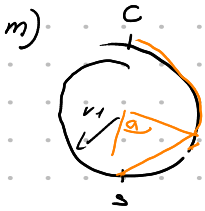
$\min\{x_1 + x_2 + \dots + x_n \mid x_1 \cdot x_2 \cdot \dots \cdot x_n > 0,\ x_i \in \{-1;1\}\}$

n: sudé $\to \arg\min = \{(-1,-1,\dots,-1)\}$ $\min = -n$

n: liché $\to \arg\min = \{(1,-1,\dots,-1), (-1,1,\dots,-1), \dots, (-1,-1,\dots,1)\}$,

$\min = 2-n$

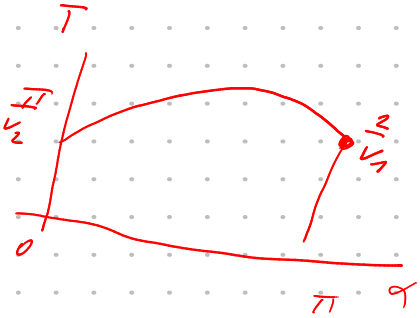
m)



plave v_1

běhá v_2

$\alpha \in \langle 0; \pi \rangle$

$\frac{d_1}{2} = \sin\frac{\alpha}{2} \rightarrow d_1 = 2\sin\frac{\alpha}{2}$, $d_2 = \pi - \alpha$

$$\min\left\{ \frac{2\sin\alpha/2}{v_1} + \frac{\pi-\alpha}{v_2} \,\middle|\, \alpha \in \langle 0; \pi \rangle \right\}$$